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Table 3.1 Stages of Production

Units of variable factor (L)	Total Product (TP_L)	Marginal Product (MP_L)	Average Product (AP_L)	Stages
1	2	2	2	I
2	6	4	3	
3	12	6	4	
4	16	4	4	II
5	18	2	3.6	
6	18	0	3	III
7	16	-2	2.28	

are three stages of production. Though total product increases steadily at first instant, constant at the maximum point and then diminishes, it is always positive for ever. While total product increases, marginal product increases up to a point and then decreases. Total product increases up to the point where the marginal product is zero. When total product tends to diminish marginal product becomes negative.

In diagram 3.1, the number of workers is measured on X axis while TP_L, AP_L and MP_L are denoted on Y axis. The diagram explains the three stages of production as given in the above table.

Stage I

In the first stage MP_L increases up to third labourer and it is higher than the average product, so that total product is increasing at an increasing rate. The tendency of total product to increase at an increasing rate stops at the point A and it begins to increase at a decreasing rate. This point is known as ***'Point of Inflexion'***.

Stage II

In the second stage, MP_L decreases up to sixth unit of labour where MP_L curve intersects the X-axis. At fourth unit of labor $MP_L = AP_L$. After this, MP_L curve is lower than the AP_L. TP_L increases at a decreasing rate.

Stage III

Third stage of production shows that the sixth unit of labour is marked by negative MP_L, the AP_L continues to fall but remains positive. After the sixth unit, TP_L declines with the employment of more units of variable factor, labour.